

Molecular Property Prediction: A Comparative Study of Graph Neural Networks and 3D Geometric Models on QM9

Bilal Malik

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1 Abstract

Predicting quantum chemical properties of molecules from their structure is a fundamental challenge in computational chemistry with direct applications in drug discovery and materials design. Classical machine learning approaches rely on fixed molecular fingerprints that discard both graph topology and three-dimensional geometry, imposing an information bottleneck that limits predictive accuracy. In this work, I conduct a systematic comparison of five models of increasing structural awareness on the QM9 benchmark dataset [1]: a Morgan fingerprint-based multilayer perceptron (MLP), Graph Convolutional Network (GCN) [2], Graph Isomorphism Network (GIN) [3], Graph Attention Network (GATv2) [4], and the 3D geometric model SchNet [5]. We evaluate all models on HOMO-LUMO gap prediction and extend the best-performing models to four additional quantum properties. Our results demonstrate that topology-based models plateau in the 0.11–0.15 eV MAE range regardless of architectural complexity, while SchNet achieves 0.0667 eV, which represents a 55% reduction over the MLP baseline. Through ablation studies, per-molecule error analysis, and attention weight visualization, we show that SchNet’s advantage is physically grounded: the magnitude of improvement scales with the geometric dependence of each target property, reaching 84.1% for the dipole moment and 41.5% for the HOMO-LUMO gap. These findings confirm that the primary bottleneck for molecular property prediction is not architectural expressiveness but the richness of the input molecular representation.

2 Introduction

The ability to accurately predict molecular properties from structure is central to modern computational chemistry. In drug discovery, properties such as frontier orbital energies and dipole moments determine how a candidate molecule interacts with biological targets [1]. In materials science, quantum mechanical properties govern conductivity, reactivity, and stability. Computing these properties from first principles using density functional theory (DFT) is accurate but computationally prohibitive at scale, motivating the development of machine learning surrogates that can predict properties orders of magnitude faster.

Early machine learning approaches represent molecules as fixed-length vectors known as molecular fingerprints, notably Morgan fingerprints [6], which encode the presence or absence of local substructure patterns in a binary vector. While effective for many tasks, fingerprint-based methods have a fundamental limitation: they discard the three-dimensional geometry of the molecule entirely, and represent connectivity only through handcrafted substructure patterns rather than learned

representations. For quantum properties that depend on the full 3D arrangement of atoms, such as the dipole moment or frontier orbital energies, this information loss imposes a hard ceiling on predictive performance.

Graph Neural Networks (GNNs) offer a natural alternative by representing molecules as graphs, where atoms are nodes and chemical bonds are edges, and learning representations directly from this structure through iterative message passing [7]. This eliminates the need for handcrafted descriptors and allows the model to learn which structural patterns are predictively relevant. Several GNN architectures have been proposed with varying expressive power: GCN [2] uses a fixed normalized aggregation, GIN [3] replaces it with a learned aggregation function proven to be maximally expressive among message passing networks, and GAT [4] introduces attention weights that allow the model to focus on more relevant neighbors.

However, all topology-based GNNs share a common limitation, they operate on the bond graph, which encodes which atoms are connected but not where they are in three-dimensional space. For quantum properties determined by the 3D electron density of the molecule, this representation is still insufficient. SchNet [5] addresses this by constructing edges from spatial proximity rather than bond connectivity, encoding interatomic distances through Gaussian radial basis functions. This makes the model rotation-invariant and gives it direct access to the 3D geometric information that governs quantum mechanical properties.

In this paper, I systematically compare these five approaches on the QM9 dataset [1]; 134,000 small organic molecules labeled with 19 quantum chemical properties computed from DFT. The comparison is designed to answer three questions: (1) How much does access to graph topology improve over flat fingerprint representations? (2) Does architectural complexity within topology-based GNNs, attention, jumping knowledge, yield meaningful gains? (3) How large is the advantage of 3D geometric models over bond-graph models, and does it generalize across different types of quantum properties?

Beyond the primary benchmark, I also conduct hyperparameter ablation studies to characterize over-smoothing behavior across architectures, perform per-molecule error analysis to identify structural failure modes, evaluate multi-target generalization across five QM9 properties, and analyze GAT attention weights for chemical interpretability. Together, these analyses provide a comprehensive picture of where each architectural family succeeds, where it fails, and why.

3 Dataset and Problem Setup

3.1 The QM9 Dataset

We conduct all experiments on the QM9 dataset [1], a benchmark collection of 134,000 small organic molecules composed of up to nine heavy atoms drawn from carbon, hydrogen, nitrogen, oxygen, and fluorine. Each molecule is annotated with 19 quantum chemical properties computed from DFT at the B3LYP/6-31G(2df,p) level of theory, covering electronic, thermodynamic, and vibrational properties. QM9 has become the standard benchmark for molecular property prediction, analogous to the role MNIST plays in image classification.

3.2 Target Property

We select the HOMO-LUMO gap as our primary prediction target. The gap is defined as the difference in energy between the highest occupied molecular orbital (HOMO) and the lowest

unoccupied molecular orbital (LUMO), measured in electron volts (eV). It governs a molecule’s optical absorption, chemical reactivity, and charge transfer behavior, making it one of the most chemically significant properties in the dataset. As shown in Figure 1, the gap values in QM9 span a wide range from approximately 2 to 11 eV, with a mean of 6.44 eV and standard deviation of 1.27 eV, reflecting the structural diversity of the dataset.

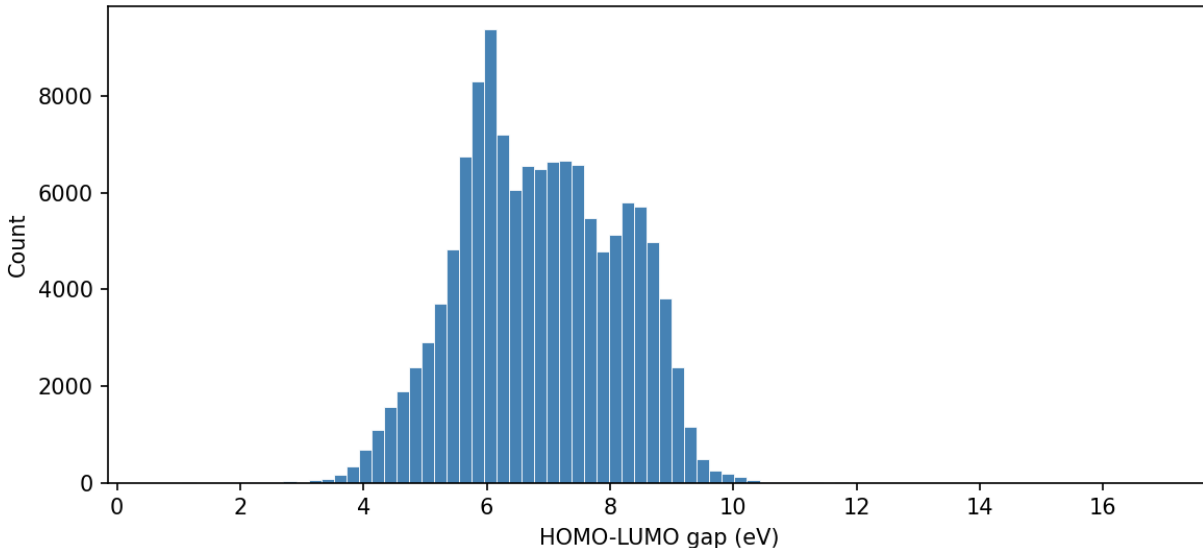


Figure 1: Distribution of HOMO-LUMO gap values across the QM9 dataset.

In Section 6.5, we extend our evaluation to four additional QM9 properties to assess whether findings on the HOMO-LUMO gap generalize across different physical origins.

3.3 Graph Construction

Each molecule is represented as an undirected graph $G = (V, E)$ where nodes V correspond to atoms and edges E correspond to chemical bonds. Node features $\mathbf{x}_i \in \mathbb{R}^{11}$ encode atomic properties including atomic number, hybridization state (sp, sp2, sp3), aromaticity, and the number of attached hydrogen atoms. Edge features $\mathbf{e}_{ij} \in \mathbb{R}^4$ encode bond type as a one-hot vector over single, double, triple, and aromatic bonds. For SchNet, the graph is discarded entirely and replaced by a radius graph constructed from the 3D atomic coordinates $\mathbf{r}_i \in \mathbb{R}^3$ stored in the dataset, connecting all atom pairs within a cutoff distance of 10 Angstroms.

3.4 Data Splits

The dataset is partitioned into training (110,000), validation (10,000), and test (10,831) splits using a fixed random seed for reproducibility. All models are trained on the same split, evaluated on the same test set, and compared under identical conditions. Target values are normalized using the mean and standard deviation computed from the training split only, and denormalized before computing evaluation metrics to prevent data leakage.

3.5 Evaluation Metric

All models are evaluated using mean absolute error (MAE) on the test set, reported in the original units of each target property. MAE is preferred over root mean squared error for this task as it

is less sensitive to outlier predictions and is the standard metric for QM9 benchmarking in the literature [3], [5].

4 Models

We compare five models of increasing structural awareness, progressing from a flat fingerprint baseline to a fully 3D geometric model.

4.1 MLP Baseline

Our baseline represents each molecule as a 2048-bit Morgan fingerprint [6] with radius 2, computed using RDKit [8]. Morgan fingerprints are circular fingerprints that encode the presence of sub-structure patterns within a fixed bond-radius neighborhood of each atom. While widely used in cheminformatics, they discard all information about global topology and three-dimensional geometry. We train a four-layer MLP with hidden dimensions [1024, 512, 256], batch normalization, ReLU activations, and dropout of 0.2 on top of these fingerprints.

4.2 Graph Convolutional Network

The GCN [2] operates directly on the molecular graph through iterative neighborhood aggregation. At each layer l , the representation of node i is updated as:

$$\mathbf{h}_i^{(l+1)} = \text{ReLU} \left(\sum_{j \in \mathcal{N}(i) \cup \{i\}} \frac{1}{\sqrt{d_i d_j}} \mathbf{W}^{(l)} \mathbf{h}_j^{(l)} \right)$$

where d_i denotes the degree of node i and $\mathbf{W}^{(l)}$ is a learned weight matrix. After L rounds of message passing, a global mean pooling operation aggregates all node representations into a fixed-size graph-level embedding, which is passed to an MLP prediction head. We use $L = 4$ layers with hidden dimension 128.

4.3 Graph Isomorphism Network

GIN [3] replaces GCN’s fixed normalized aggregation with a learned aggregation function, making it maximally expressive among message passing networks in the sense that it can distinguish any pair of non-isomorphic graphs that the Weisfeiler-Leman graph isomorphism test can distinguish. The update rule is:

$$\mathbf{h}_i^{(l+1)} = \text{MLP}^{(l)} \left((1 + \epsilon^{(l)}) \mathbf{h}_i^{(l)} + \sum_{j \in \mathcal{N}(i)} \mathbf{h}_j^{(l)} \right)$$

where ϵ is a learnable parameter. We additionally employ jumping knowledge [9] by concatenating the graph-level pooled representations from all L layers before the prediction head, allowing the model to capture both local and global molecular structure simultaneously.

4.4 Graph Attention Network

GATv2 [4] extends GCN by replacing the fixed aggregation weights with dynamic attention coefficients computed between every pair of connected atoms:

$$\alpha_{ij} = \frac{\exp(\mathbf{a}^\top \text{LeakyReLU}(\mathbf{W}[\mathbf{h}_i \parallel \mathbf{h}_j]))}{\sum_{k \in \mathcal{N}(i)} \exp(\mathbf{a}^\top \text{LeakyReLU}(\mathbf{W}[\mathbf{h}_i \parallel \mathbf{h}_k]))}$$

We use GATv2 rather than the original GAT [1] as it corrects a theoretical limitation in which the original formulation computes static rather than dynamic attention. We use 2 attention heads with hidden dimension 128 and a learning rate of 3×10^{-4} , which was found necessary for stable convergence.

4.5 SchNet

SchNet [5] departs fundamentally from the bond-graph models by constructing a radius graph from 3D atomic coordinates and encoding interatomic distances through a bank of Gaussian radial basis functions:

$$e_k(r_{ij}) = \exp\left(-\gamma \|r_{ij} - \mu_k\|^2\right)$$

where μ_k are evenly spaced centers and γ controls the width. These distance features are used to compute continuous-filter convolutions that update atomic representations based on physically meaningful spatial interactions rather than topological connectivity. This makes SchNet rotation-invariant and gives it direct access to the 3D structural information that governs quantum mechanical properties. We use 6 interaction blocks, 50 Gaussian basis functions, and a cutoff radius of 10 Angstroms.

5 Experimental Setup

All models are trained with the Adam optimizer [10] with weight decay 10^{-5} . Bond-graph models (GCN, GIN, GAT) use a cosine annealing learning rate schedule over 100 epochs with an initial learning rate of 10^{-3} , except GAT which uses 3×10^{-4} for training stability. SchNet uses a ReduceLROnPlateau scheduler with reduction factor 0.5 and patience 10, which was found to produce more stable convergence than cosine annealing for the geometry-based model. All models use a hidden dimension of 128, 4 message passing layers (6 interaction blocks for SchNet), and a batch size of 64. Gradient clipping at norm 1.0 is applied to all models.

Bond-graph models are trained locally on CPU. SchNet and ablation variants are trained on a Kaggle P100 GPU using mixed-precision training (torch.amp.autocast) for efficiency. All ablation experiments use 50 training epochs with a ReduceLROnPlateau scheduler to ensure fair comparison across architectures under an identical training budget.

Models are evaluated on the test set using the checkpoint that achieved the lowest validation MAE during training. All experiments use a fixed random seed of 42 for dataset splitting and weight initialization to ensure reproducibility.

6 Benchmark Results

6.1 Quantitative Comparison

Table 1 presents the test MAE for all five models on the HOMO-LUMO gap prediction task. The MLP baseline achieves 0.1467 eV, establishing the performance ceiling for flat fingerprint representations. Among topology-based GNNs, GIN achieves the best result at 0.1140 eV, followed by GCN at 0.1369 eV and GAT at 0.1453 eV. SchNet achieves 0.0667 eV, a 55% reduction over the MLP baseline and a 41% reduction over the best topology-based model.

Table 1: Test MAE on HOMO-LUMO gap prediction for all five models.

Model	Representation	Test MAE (eV)	vs. Baseline
MLP	Morgan fingerprints	0.1467	—
GAT	Graph attention	0.1453	-1.0%
GCN	Graph topology	0.1369	-6.7%
GIN	Expressive topology	0.1140	-22.3%
SchNet	3D geometry	0.0667	-54.5%

The most striking observation is that all topology-based GNNs cluster in the 0.11–0.15 eV range, regardless of architectural complexity. The difference between the weakest topology-based model (GAT, 0.1453 eV) and the strongest (GIN, 0.1140 eV) is only 0.0313 eV, smaller than SchNet’s improvement over GIN alone (0.0473 eV). This suggests that the primary bottleneck for bond-graph models is not architectural expressiveness but the information content of the bond-graph representation itself.

6.2 Training Dynamics

Figure 2 presents the validation MAE curves for all four GNN models over 100 training epochs, alongside the MLP baseline as a reference.

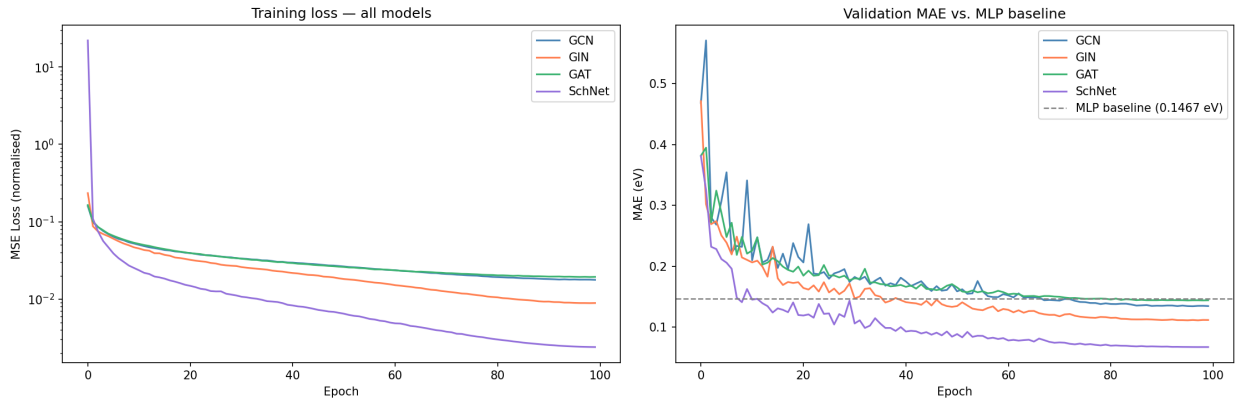


Figure 2: Validation MAE over 100 training epochs for all models. The dashed line indicates the MLP baseline (0.1467 eV).

GCN and GAT show early rapid improvement followed by a plateau, with validation MAE largely stabilizing by epoch 60. GIN converges more gradually but continues improving through epoch 100,

consistent with its jumping knowledge architecture capturing multi-scale structural information that requires more training to consolidate. SchNet exhibits the most gradual convergence, with a characteristic early oscillation between epochs 1 and 40 as the model learns which distance ranges carry predictive signal, before stabilizing into steady improvement. Notably, SchNet’s validation MAE at epoch 100 (0.0674 eV) had not fully plateaued, suggesting that additional training epochs would yield further improvement.

6.3 Error Distributions

Beyond mean MAE, the distribution of per-molecule errors reveals important differences in model behavior. Figure 3 presents error histograms for all four GNN models on the test set.

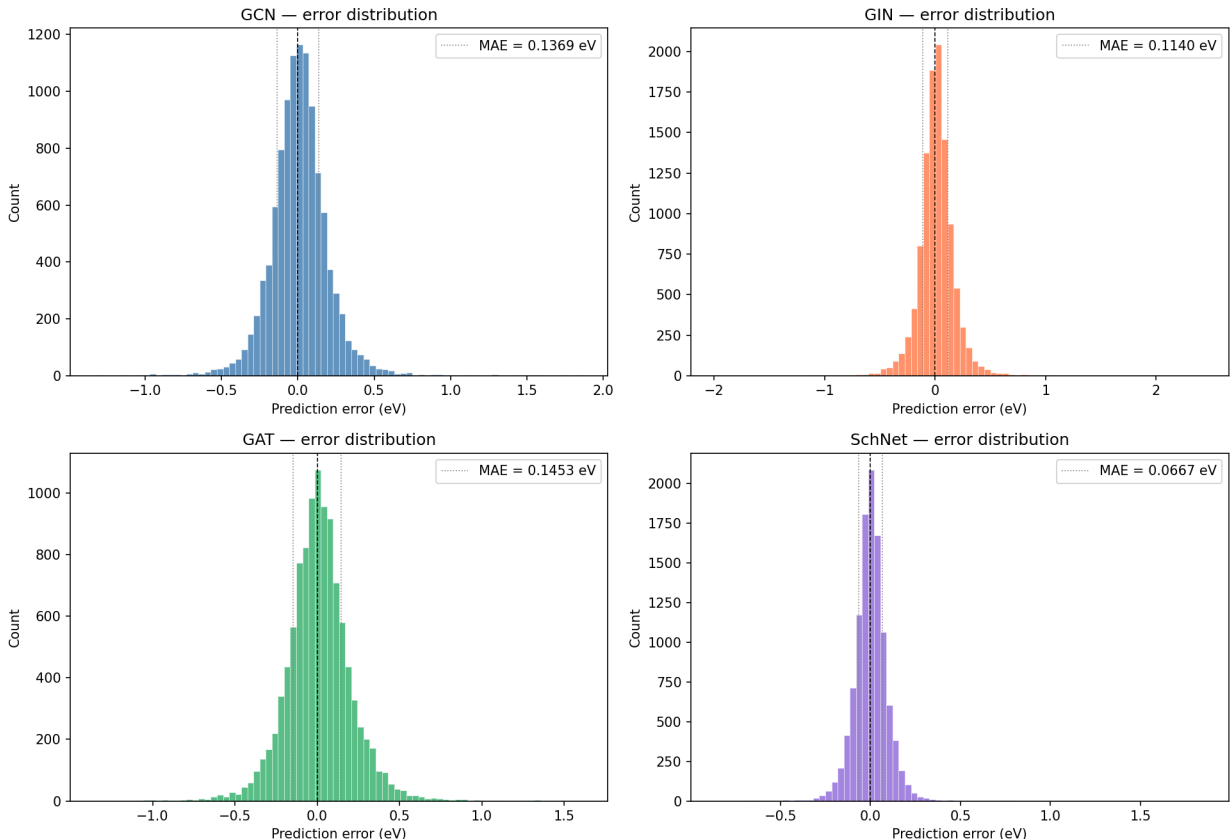


Figure 3: Per-molecule absolute error distributions on the test set. Dotted vertical lines indicate the mean absolute error.

Table 2 summarizes the key distributional statistics.

Table 2: Error distribution statistics (eV) for all GNN models on the test set.

Model	Mean	Median	95th Pct.	Max
GCN	0.1369	0.1019	0.3813	1.8693
GIN	0.1140	0.0813	0.3248	2.4340
GAT	0.1453	0.1084	0.3960	1.6172

Model	Mean	Median	95th Pct.	Max
SchNet	0.0667	0.0490	0.1821	1.8540

Two observations are notable. First, SchNet’s 95th percentile error (0.1821 eV) is lower than the mean MAE of GCN, GAT, and nearly GIN, meaning SchNet’s worst 5% of predictions are still more accurate than the average prediction of the bond-graph models. Second, GIN has the highest maximum error (2.434 eV) despite its best mean MAE among bond-graph models. This pathological tail behavior is consistent with the embedding space collapse discussed in Section 6.4, GIN’s over-compressed representations occasionally fail catastrophically on structurally unusual molecules that fall outside its learned clusters.

6.4 Learned Molecular Representations

To assess the quality of learned representations beyond predictive performance, we extract graph-level embeddings from each GNN model before the prediction head and project them to two dimensions using UMAP [11]. Figure 4 presents the resulting embedding spaces for all four models, colored by true HOMO-LUMO gap value.

The four embedding spaces reveal qualitatively different representation strategies. GCN produces a fragmented space of disconnected islands, grouping molecules by simple structural patterns but without global chemical organization. Molecules with similar gap values are scattered across separate clusters rather than forming a continuous manifold. GAT produces the least organized space, points are spread diffusely with poor color coherence, consistent with the attention mechanism failing to learn a meaningful chemical representation on small molecular graphs.

GIN presents an unusual case: the embedding space collapses into a small number of very tight, well-separated clusters with large empty regions between them. While GIN achieves the best MAE among bond-graph models, its discrete clustering suggests the jumping knowledge architecture produces highly compressed, discriminative representations that partition molecules into bins rather than placing them on a smooth chemical manifold. This over-compression likely explains GIN’s pathological maximum error, molecules that fall between clusters have no nearby reference points in the learned space.

SchNet’s embedding space stands in contrast to all three bond-graph models. The projection forms a single connected manifold with a smooth, continuous color gradient from low-gap molecules (purple, bottom) to high-gap molecules (yellow, top). This structured organization indicates that SchNet has learned a representation that reflects the underlying quantum chemical relationships between molecules, not merely a pattern-matching function, but something closer to a learned chemical space.

7 In-Depth Analysis

7.1 Hyperparameter Ablation: Message Passing Depth

A central concern in GNN design is the relationship between network depth and performance. As message passing depth increases, node representations aggregate information from progressively larger neighborhoods. Beyond a certain depth, representations of distant nodes become indistinguishable, a phenomenon known as over-smoothing [12], and performance degrades. To characterize this

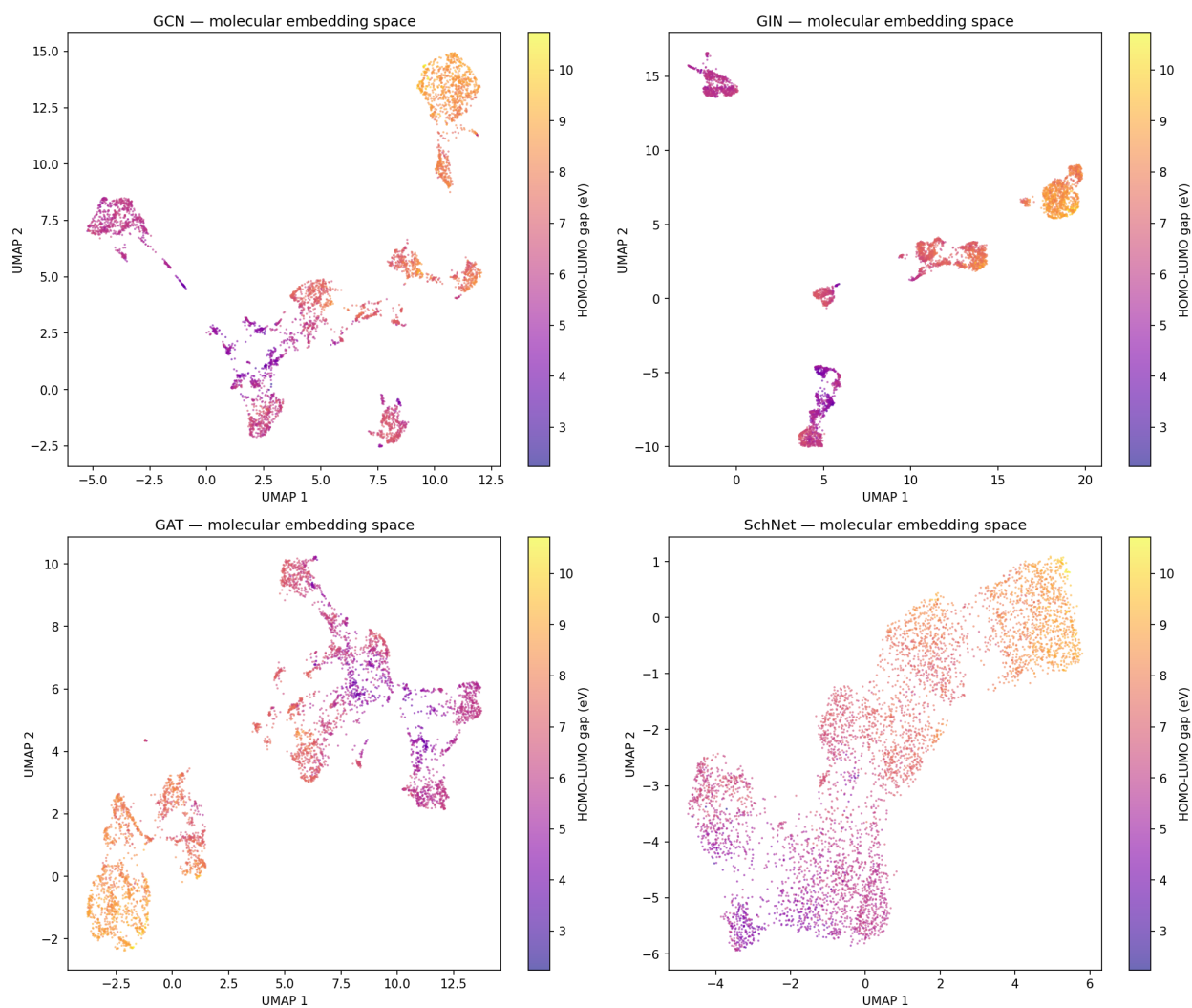


Figure 4: UMAP projections of learned molecular embeddings for all four GNN models, colored by HOMO-LUMO gap value (eV). A well-learned chemical space should show a smooth, continuous color gradient.

behavior empirically, we train GCN, GIN, and GAT with 1 to 5 message passing layers, keeping all other hyperparameters fixed. Figure 5 presents the test MAE as a function of depth.

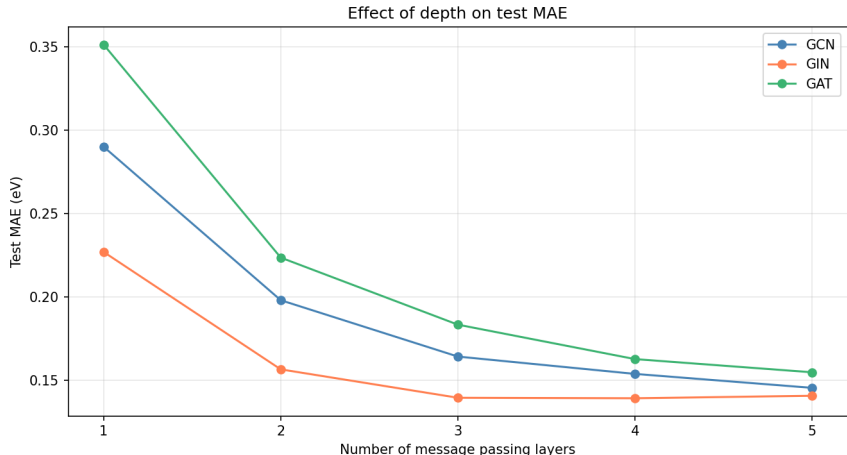


Figure 5: Test MAE as a function of message passing depth for GCN, GIN, and GAT. All variants trained for 50 epochs under identical conditions.

All three models improve monotonically from 1 to 5 layers with no performance degradation, the over-smoothing regime is never reached within this range. This result is not a failure of the experiment but a physically meaningful finding: QM9 molecules contain a median of 9 heavy atoms, and with such small graphs, even 5 layers of message passing cannot fully homogenize node representations the way it would on large citation networks or social graphs. The structural diameter of most QM9 molecules is smaller than the depth at which over-smoothing typically activates.

GIN shows the earliest saturation, with MAE improving by only 0.0015 eV between layers 4 and 5 (0.1394 vs 0.1409 eV), compared to GCN’s continued improvement of 0.0083 eV over the same range. This is consistent with GIN’s jumping knowledge architecture, by concatenating pooled representations from all layers, GIN extracts multi-scale structural information efficiently at shallow depths, leaving less additional signal for deeper layers to capture.

7.2 Hyperparameter Ablation: SchNet Interaction Blocks

We train SchNet with 3, 4, 6, and 8 interaction blocks to characterize how depth affects the geometry-aware model. Unlike the bond-graph models, SchNet exhibits a non-monotonic depth response, with optimal performance at 4 blocks and degraded performance at 6 and 8 blocks (Table 3).

Table 3: SchNet test MAE as a function of interaction block depth. Bold indicates best performance.

Interaction blocks	Test MAE (eV)
3	0.0958
4	0.0777
6	0.0870
8	0.0867

The degradation beyond 4 blocks reflects a geometric analog of over-smoothing. With more

interaction blocks, each atom iteratively aggregates information from neighbors of its neighbors within the radius graph. At sufficient depth, atoms effectively see contributions from every other atom in the molecule regardless of their true 3D distance, diluting the local geometric signal that makes SchNet effective. This mechanism differs from topological over-smoothing in an important way: in bond-graph models, over-smoothing is driven by graph connectivity structure; in SchNet, it is driven by the iterative propagation of distance-encoded features beyond their physically meaningful range.

7.3 Hyperparameter Ablation: SchNet Cutoff Radius

The cutoff radius determines which atom pairs contribute to SchNet’s radius graph. Too small a cutoff starves the model of neighbor information; too large a cutoff introduces noisy long-range interactions. We train SchNet with cutoff values of 5.0, 7.5, 10.0, and 12.0 Angstroms.

Table 4: SchNet test MAE as a function of cutoff radius. Bold indicates best performance.

Cutoff (Angstroms)	Test MAE (eV)
5.0	0.1177
7.5	0.0772
10.0	0.0772
12.0	0.0957

Three distinct performance regimes emerge. Below 7.5 Angstroms, the cutoff is too restrictive, at 5.0 Angstroms, SchNet performs only marginally better than GIN (0.1177 vs 0.1140 eV), as many physically relevant atomic interactions fall outside the radius graph. Between 7.5 and 10.0 Angstroms, performance is identical (0.0772 eV), indicating that the physically meaningful interaction range for HOMO-LUMO gap prediction is fully captured within 7.5 Angstroms. This corresponds approximately to the van der Waals interaction range for the light organic molecules comprising QM9. Beyond 10.0 Angstroms, performance degrades as the radius graph approaches full connectivity, and the distance-encoded edge features lose discriminative power in the presence of many weak long-range interactions.

7.4 Per-Molecule Error Analysis

7.4.1 Error vs. Molecular Size

Figure 6 presents mean absolute error as a function of atom count for each model. Contrary to the intuitive expectation that larger molecules are harder to predict, no model shows a systematic increase in error with molecule size. Errors are modestly elevated for the smallest molecules (5 to 7 atoms) and relatively flat across medium and large molecules.

The elevated error for small molecules reflects the sparsity of their graph representations. A 5-atom molecule provides each GNN node with at most 2 to 3 neighbors to aggregate from, limiting the structural context available for message passing. SchNet is comparatively less affected by small molecule size, its error curve is notably flatter than the bond-graph models, consistent with 3D pairwise distances providing a richer signal than sparse graph topology even when the number of atoms is small.

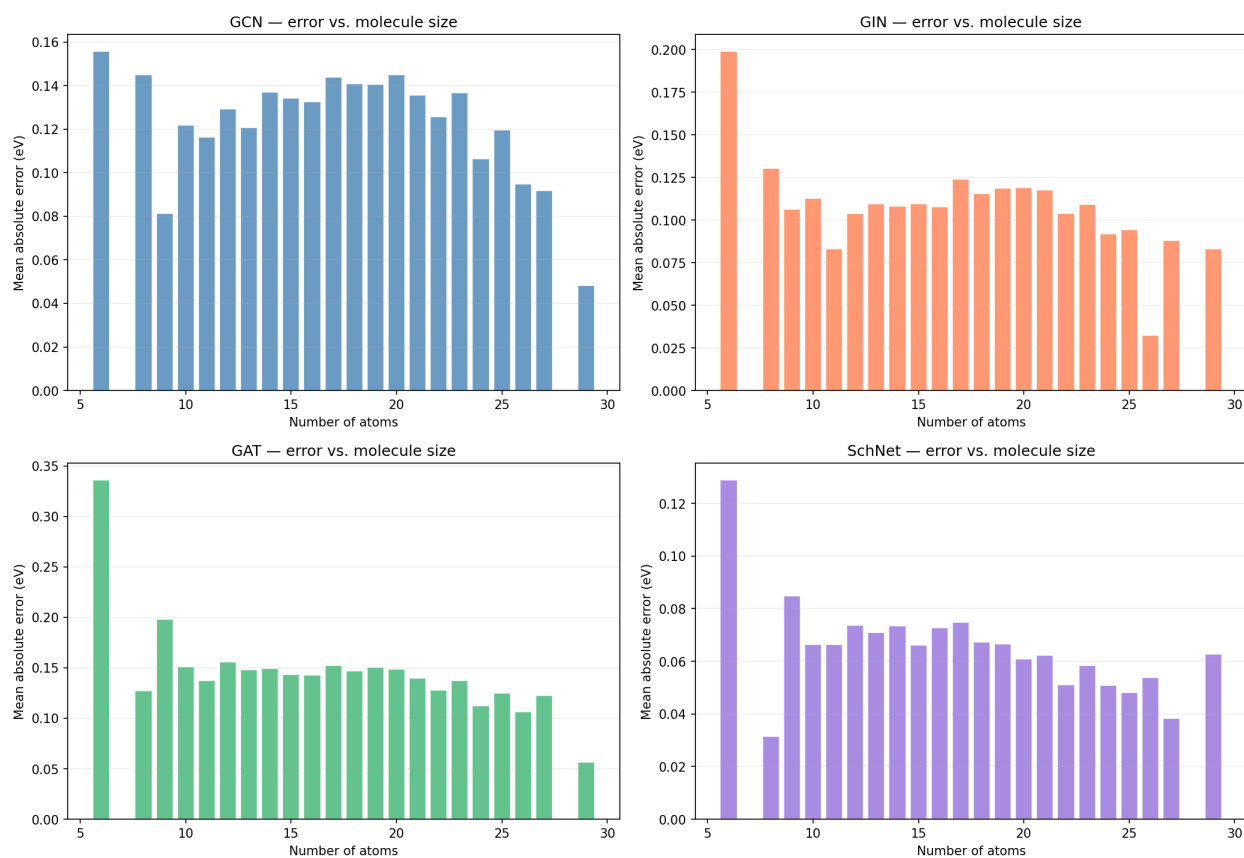


Figure 6: Mean absolute error as a function of molecule size (number of atoms) for all four GNN models.

7.4.2 Error vs. Structural Properties

Figure 7 presents mean absolute error stratified by ring count, aromaticity, and heteroatom presence.

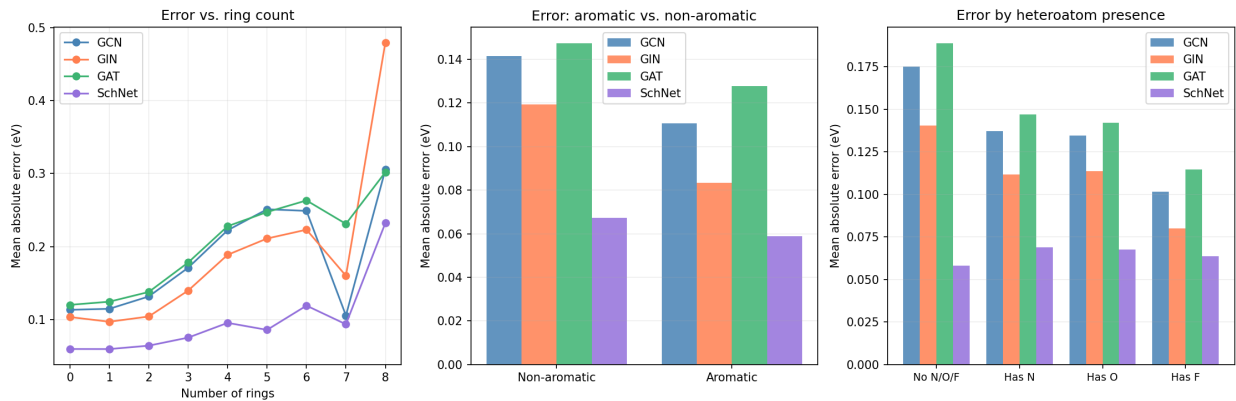


Figure 7: Mean absolute error stratified by structural properties: ring count (left), aromaticity (center), and heteroatom presence (right).

Error increases monotonically with ring count for all models, with a consistent ranking across the full range. This reflects the increasing geometric complexity of ring systems, polycyclic and bridged ring compounds impose unusual bond angles and strained geometries that are underrepresented in the training distribution. SchNet’s error curve is consistently lower and flatter than the bond-graph models, but it does not eliminate the ring-count dependence, indicating that strained polycyclic systems remain challenging even for geometry-aware models.

All four models perform better on aromatic molecules than non-aromatic ones. Aromatic systems have regular, planar geometries and HOMO-LUMO gaps that cluster in a narrower, more predictable range compared to the structural diversity of non-aromatic molecules. The improvement is largest for GIN (29%) and GCN (22%), likely because aromatic bond type is explicitly encoded in the edge features and the model can leverage this signal directly.

For heteroatom presence, fluorine-containing molecules produce the lowest errors across all models. Fluorine’s strong electron-withdrawing effect dominates the HOMO-LUMO gap in a predictable way, providing a strong and consistent signal regardless of the rest of the molecular structure. Pure hydrocarbon molecules show the highest errors, consistent with the observation that they tend to be structurally complex cage and bicyclic compounds within QM9.

7.4.3 Worst-Case Predictions

Figure 8 and Figure 9 present the 12 highest-error molecules for GCN and SchNet respectively.

Both models’ worst predictions are dominated by strained bicyclic and cage compounds, structures with unusual bond angles and highly constrained 3D geometries that are rare in the training distribution. The worst-case molecule for GCN (MAE: 1.869 eV) and for SchNet (MAE: 1.854 eV) is the same epoxide-amine bicyclic compound, confirming that these failures are driven by the chemistry of the molecule rather than by architectural limitations. These structures represent genuine out-of-distribution cases for any model trained on QM9.

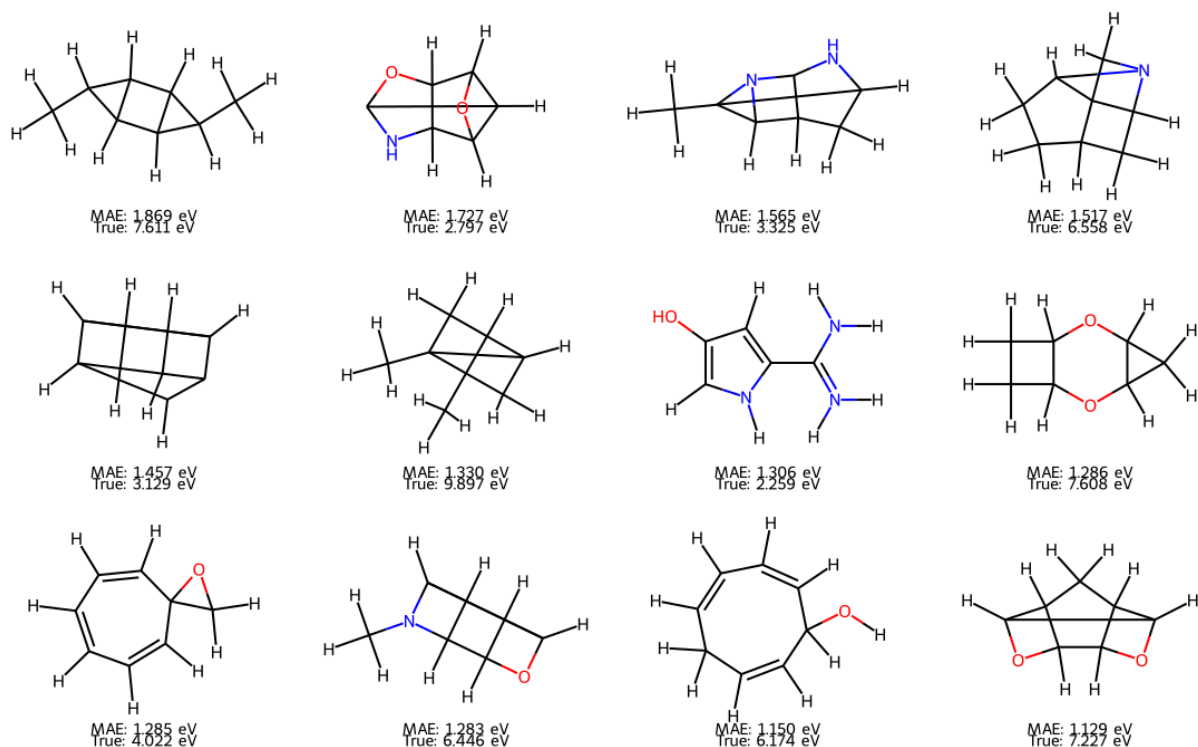


Figure 8: The 12 highest-error molecules for GCN on the test set, with absolute error and true gap value annotated.

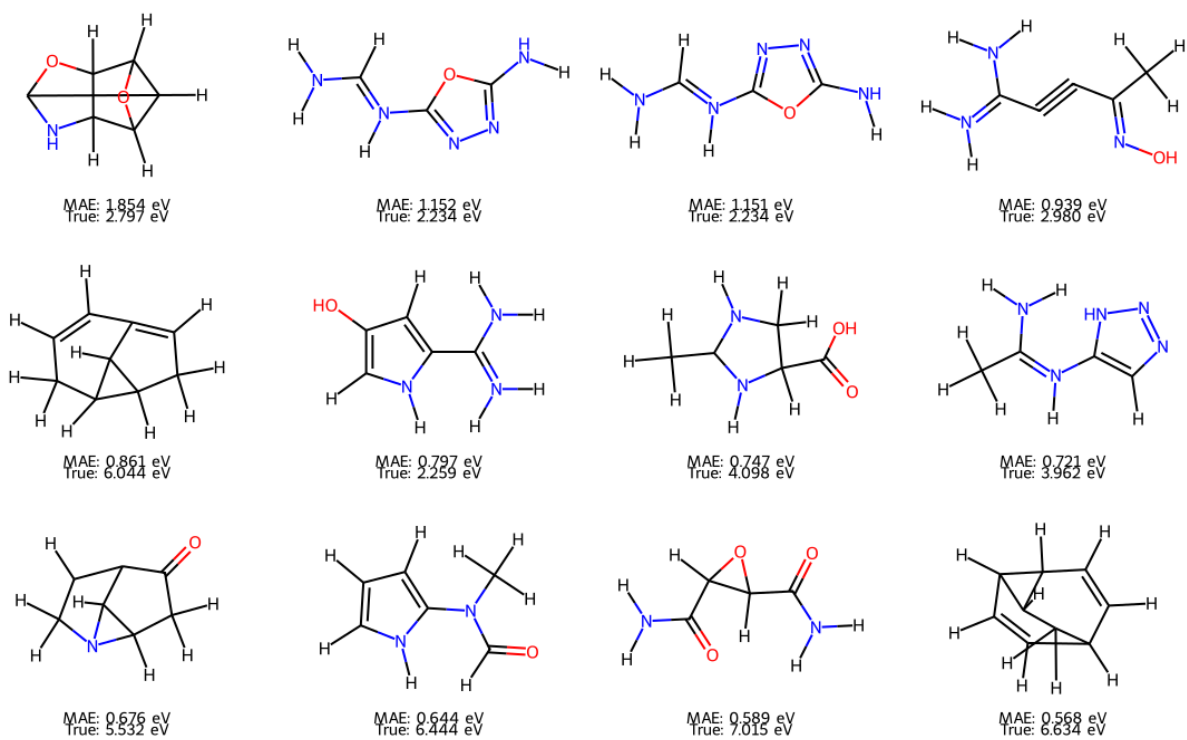


Figure 9: The 12 highest-error molecules for SchNet on the test set.

7.4.4 Cross-Model Error Correlation

Figure 10 presents pairwise scatter plots of per-molecule absolute errors between model pairs.

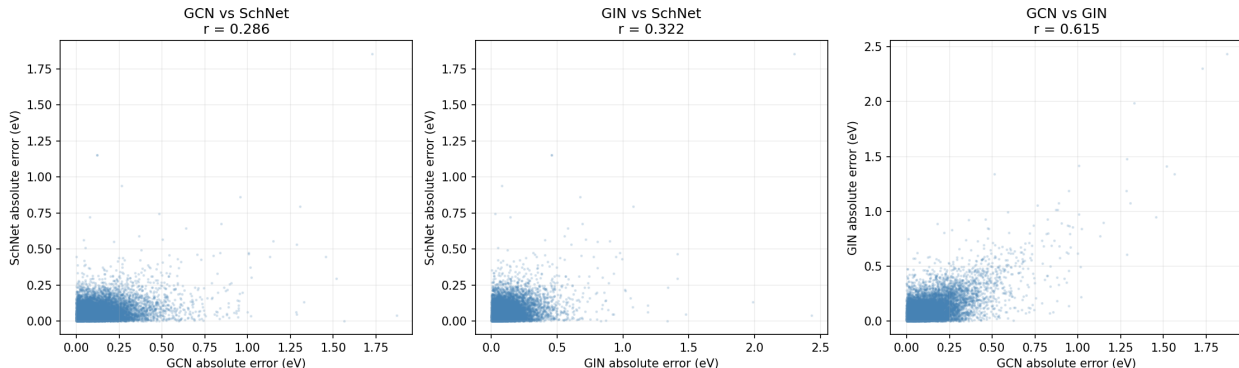


Figure 10: Pairwise scatter plots of per-molecule absolute errors. Pearson correlation coefficients are shown above each plot.

The correlation between GCN and GIN errors ($r = 0.615$) is substantially higher than between either bond-graph model and SchNet ($r = 0.286$ and $r = 0.322$ respectively). This confirms that topology-based models share failure modes driven by their common information source, bond connectivity, while SchNet’s failures are largely independent. When SchNet makes a large error, GCN and GIN are often accurate on the same molecule, and vice versa. The two architectural families have genuinely different blind spots, suggesting that an ensemble of topology-based and geometry-based models could outperform either family alone.

7.5 Multi-Target Evaluation

To assess whether SchNet’s advantage over GIN generalizes beyond the HOMO-LUMO gap, we train both models on four additional QM9 properties spanning different physical origins. Table 5 presents the results.

Table 5: GIN vs. SchNet test MAE across five QM9 properties. Improvement is SchNet’s percentage reduction over GIN.

Property	Physical origin	GIN MAE	SchNet MAE	Improvement
gap (eV)	Orbital energy difference	0.1140	0.0667	41.5%
homo (eV)	Orbital energy	0.0871	0.0428	50.9%
Cv (cal/mol/K)	Thermodynamic	0.1558	0.0796	48.9%
alpha (a0 ³)	Polarizability	0.3902	0.2042	47.7%
mu (D)	Dipole moment	0.4065	0.0645	84.1%

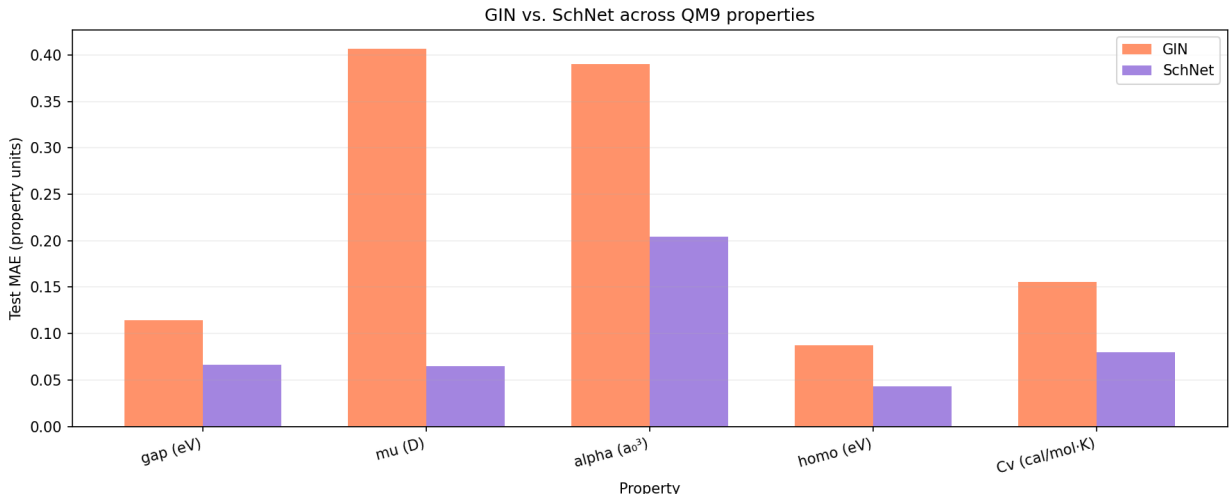


Figure 11: GIN vs. SchNet MAE across five QM9 properties (left) and SchNet percentage improvement over GIN per property (right).

SchNet outperforms GIN on all five properties, but the magnitude of improvement varies systematically with the geometric dependence of each property. The dipole moment shows the largest improvement at 84.1%, this is physically expected, as the dipole moment is defined as the vector sum of charge separations in 3D space and cannot be computed without knowing atomic positions. GIN’s MAE of 0.4065 D for this property reflects the near-complete absence of relevant signal in bond topology alone.

The HOMO energy (50.9%) and heat capacity (48.9%) show comparable improvements, followed by polarizability (47.7%) and the HOMO-LUMO gap (41.5%). The gap shows the smallest SchNet advantage despite being a quantum electronic property, a likely consequence of error cancellation when subtracting HOMO and LUMO energies, both of which individually show larger SchNet advantages.

Critically, SchNet’s 48.9% improvement on heat capacity refutes the alternative hypothesis that SchNet’s advantage is limited to electronic properties. Heat capacity depends on vibrational modes determined by the 3D potential energy surface, confirming that SchNet’s advantage extends to thermodynamic quantities where 3D atomic positions govern the physics. The systematic scaling of improvement with geometric dependence provides strong evidence that SchNet’s superior performance reflects genuine physical understanding rather than a model capacity effect.

7.6 GAT Attention Interpretability

A key motivation for attention-based architectures is interpretability, the hope that learned attention weights reflect chemically meaningful patterns. We extract attention weights from all four GATv2 layers for molecules in the test set and analyze mean attention weight by bond type.

Note that the PyG QM9 dataset Kekulizes aromatic bonds into alternating single and double bonds during preprocessing, meaning aromatic bond type is not present in the edge features and could not be analyzed directly.

Figure 12 presents mean attention weights for single and double bonds across five successfully

visualized test molecules.

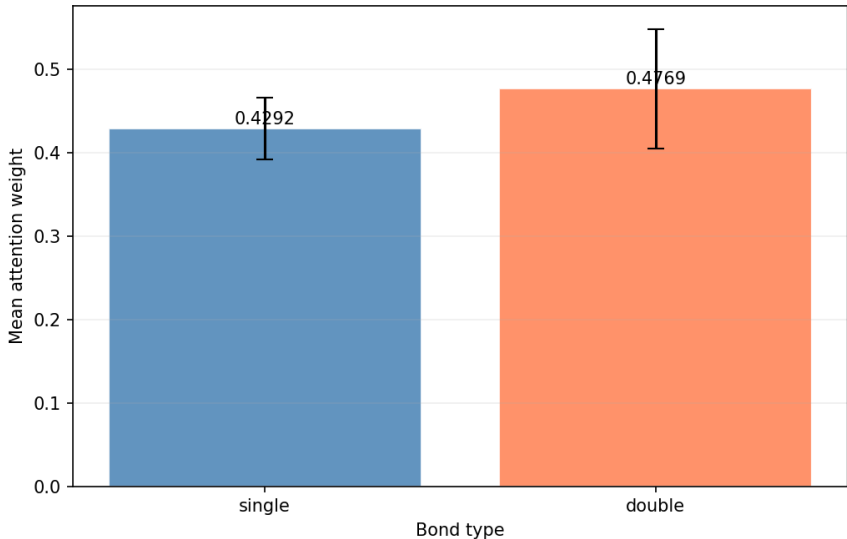


Figure 12: Mean GAT attention weight by bond type, averaged across five test molecules and all four attention layers. Error bars indicate one standard deviation.

Double bonds receive modestly higher mean attention (0.4769) than single bonds (0.4292), and this pattern holds in 3 of 4 molecules containing both bond types. However, the difference is small and the standard deviations overlap substantially, indicating that the preference is consistent but weak.

This result is interpretable in the context of GAT’s performance on QM9. The attention mechanism in GATv2 computes weights by comparing each node’s representation to its neighbors. In QM9 molecules with 5 to 15 atoms, most nodes have at most 2 to 3 bonded neighbors, which severely limits the range over which attention can differentiate. A node attending over two neighbors can only split its attention two ways, there is little room for the sharp specialization seen in larger, more heterogeneous graphs. The near-uniform attention weights observed here are therefore not a failure of the attention mechanism per se, but a consequence of the structural constraints of the data, consistent with GAT’s marginal performance advantage over GCN on this dataset.

8 Discussion

The results across both benchmark evaluation and in-depth analysis converge on three central findings that together characterize the relationship between molecular representation, architectural complexity, and predictive performance on quantum chemical properties.

The input representation is the primary bottleneck, not architectural expressiveness. All topology-based models, GCN, GIN, and GAT, plateau in the 0.11 to 0.15 eV MAE range regardless of how sophisticated their aggregation mechanism is. GIN’s learned aggregation and jumping knowledge outperform GCN’s fixed normalized sum, but the margin is modest (0.1140 vs 0.1369 eV). GAT’s dynamic attention mechanism, despite being the most architecturally complex of the three, underperforms GIN and nearly ties the MLP baseline (0.1453 vs 0.1467 eV). The consistent clustering of all three models within the same performance band suggests they share a common information ceiling imposed by bond topology, the representation itself, not the architecture, is the limiting factor. This finding has practical implications: investing in more expressive bond-graph

architectures for quantum property prediction yields diminishing returns compared to improving the molecular representation.

SchNet’s advantage is physically grounded rather than parametric. The 55% improvement of SchNet over the MLP baseline could in principle reflect simply having more parameters or a more flexible function class. The multi-target evaluation in Section 7.4 rules this out. SchNet’s improvement over GIN scales systematically with the geometric dependence of each target property: 84.1% for the dipole moment, which is by definition a 3D vector quantity; 50.9% for the HOMO energy; 48.9% for heat capacity; 47.7% for polarizability; and 41.5% for the HOMO-LUMO gap. A model with a generic capacity advantage would improve all properties roughly equally. The observed scaling with geometric dependence indicates that SchNet is extracting physically meaningful information from 3D atomic coordinates that bond-graph models cannot access.

Architectural complexity has diminishing returns on small molecular graphs. Three independent lines of evidence support this conclusion. First, the depth ablation study shows no over-smoothing up to 5 message passing layers, the graphs are simply too small for the standard failure mode of deep GNNs to activate. Second, GAT’s attention weights are nearly uniform across bond types, indicating that the attention mechanism cannot find strong differentiation signals when each node has only 2 to 3 neighbors. Third, SchNet’s optimal performance occurs at 4 interaction blocks rather than the deeper configurations, suggesting that even geometry-aware models encounter a form of over-smoothing when interaction depth exceeds what the physical scale of the molecules requires. Taken together, these observations suggest that the design principles developed for GNNs on large, heterogeneous graphs, deep architectures, attention mechanisms, multi-head aggregation, do not transfer straightforwardly to small molecular graphs where structural diversity is inherently limited.

A secondary finding of practical significance concerns the cross-model error correlation. The low Pearson correlation between SchNet and bond-graph model errors ($r = 0.286$ to 0.322) indicates that the two architectural families fail on largely different molecules. This complementarity suggests that an ensemble combining topology-based and geometry-based predictions could outperform either family alone, particularly for structurally unusual molecules where individual models are most likely to fail. Strained bicyclic and cage compounds appear consistently in the worst-case predictions of all models and represent the most promising targets for future architectural improvements.

9 Conclusion

This work presents a systematic comparison of five molecular property prediction models on the QM9 benchmark, spanning flat fingerprint representations, topology-based graph neural networks of increasing expressiveness, and a 3D geometric model. The central finding is unambiguous: access to three-dimensional atomic geometry is the single most important factor for accurate quantum chemical property prediction, contributing a 55% reduction in MAE over the fingerprint baseline and a 41% reduction over the best topology-based model. This advantage is not uniform, it scales with the geometric dependence of each target property, providing strong evidence that SchNet extracts physically meaningful structural information rather than simply benefiting from greater model capacity.

Within the family of topology-based models, architectural sophistication provides modest and inconsistent gains. GIN’s more expressive aggregation outperforms GCN meaningfully, but GAT’s attention mechanism, despite being the most complex of the three, underperforms both. Ablation

studies confirm that the standard failure modes of deep GNNs, particularly over-smoothing, do not activate within the depth ranges studied on QM9’s small molecular graphs, suggesting that the design principles for large heterogeneous graphs do not transfer directly to molecular settings.

Several directions remain open for future work. The ablation results identify that SchNet’s optimal configuration on this dataset is 4 interaction blocks and a 7.5 Angstrom cutoff radius, while the primary experiments used 6 blocks and 10 Angstroms, retraining with the optimal settings would likely yield further improvement below the reported 0.0667 eV. Extending the comparison to larger molecules beyond QM9’s nine heavy atom limit would test whether over-smoothing activates at greater depth and whether attention mechanisms become more effective when molecular graphs are larger and more structurally diverse. Finally, the low cross-model error correlation between SchNet and topology-based models motivates the development of ensemble approaches that combine geometric and topological representations, particularly for the strained polycyclic compounds that represent the consistent failure mode across all architectures studied here.

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